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Modul 1: Electrical quantities



Stand: 18. Februar 2026



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Key point: Description of physical quantities

The Students can

- ▶ specify physical quantities with measurement values and units of measurement.
- ▶ use the seven SI base units and the units derived from them, and assign them to their respective quantities
- ▶ Express numerical values using powers of ten and their names, and convert them into each other. Convert

The SI system of units

SI-Size	Formula symbol	Unit	Basis

Table: SI units and their base quantities

The SI system of units

SI-Size	Formula symbol	Unit	Basis
Time	t	Second, s	$\Delta\nu$

Table: SI units and their base quantities

One second is 9,192,631,770 times the period of the hyperfine transition. $\Delta\nu$ in a caesium atom ^{133}Cs .

$$\Delta\nu = 9.192.631.770 \text{ Hz} \quad (\text{Hyperfine transition } ^{133}\text{Cs})$$

The SI system of units

SI-Size	Formula symbol	Unit	Basis
Time	t	Second, s	$\Delta\nu$
Length	ℓ	Meter, m	c, s

Table: SI units and their base quantities

One metre is the length of the path travelled by light in a vacuum during the time of

$$t = \frac{1}{299.792.458} \text{ s travels.}$$

$$c = 299.792.458 \frac{\text{m}}{\text{s}} \quad (\text{Speed of light})$$

The SI system of units

SI-Size	Formula symbol	Unit	Basis
Time	t	Second, s	$\Delta\nu$
Length	ℓ	Meter, m	c, s
Mass	m	Kilogram, kg	h, s, m

Table: SI units and their base quantities

Since May 2019, the kilogram has been dependent on the precisely defined Planck constant: $1\text{kg} = \frac{h}{6,62607015 \cdot 10^{-34} \frac{\text{s}}{\text{m}^2}}$

$$h = 6,62607015 \cdot 10^{-34} \text{Js} \quad (\text{Planck's action quantum})$$

The SI system of units

SI-Size	Formula symbol	Unit	Basis
Time	t	Second, s	$\Delta\nu$
Length	ℓ	Meter, m	c, s
Mass	m	Kilogram, kg	h, s, m
Current value	I	Current, A	e, s

Table: SI units and their base quantities

Since 2019, the ampere has been defined by the elementary charge. One ampere is the current flow of $\frac{e}{1,602176634} \cdot 10^{-19} \frac{1}{s}$ Elementary charges per second.

$$e = 1,602176634 \cdot 10^{-19} \text{As} \quad (\text{Elementary charge})$$

The SI system of units

SI-Size	Formula symbol	Unit	Basis
Time	t	Second, s	$\Delta\nu$
Length	ℓ	Meter, m	c, s
Mass	m	Kilogram, kg	h, s, m
Current value	I	Current, A	e, s
Temperature	T	Kelvin, K	k_B, s, m, kg

Table: SI units and their base quantities

One Kelvin corresponds to a change in thermodynamic temperature that is associated with a change in thermal energy. (kT) um $1,380649 \cdot 10^{-23}$ J accompanied by.

$$k_B = 1,38064910^{-23} \frac{\text{kg} \cdot \text{m}^2}{\text{s}^2 \cdot \text{K}} \quad (\text{Boltzmann constant})$$

The SI system of units

SI-Size	Formula symbol	Unit	Basis
Time	t	Second, s	$\Delta\nu$
Length	ℓ	Meter, m	c, s
Mass	m	Kilogram, kg	h, s, m
Current value	I	Current, A	e, s
Temperature	T	Kelvin, K	k_B, s, m, kg
Amount of substance	n	Mol, mol	N_A

Table: SI units and their base quantities

One mole is the amount of substance in a system that contains $6.02214076 \cdot 10^{23}$ of a specific individual particle.

$$N_A = 6,02214076 \cdot 10^{23} \frac{1}{\text{mol}} \quad (\text{Avogadro's constant})$$

The SI system of units

SI-Size	Formula symbol	Unit	Basis
Time	t	Second, s	$\Delta\nu$
Length	ℓ	Meter, m	c, s
Mass	m	Kilogram, kg	h, s, m
Current value	I	Current, A	e, s
Temperature	T	Kelvin, K	k_B, s, m, kg
Amount of substance	n	Mol, mol	N_A
Light intensity	I_v	Candela, cd	K_{cd}, s, m, kg

Table: SI units and their base quantities

A candela is the luminous intensity of a radiation source that emits at a frequency of 540 THz and has a radiation intensity in this direction.. von $\frac{1}{683} \frac{W}{sr}$ hat. (sr = Solid angle Steradian)

$$K_{cd}(540THz) = 683 \frac{lm}{W} \quad (\text{Photometric radiation equivalent})$$

Size	Formula	Unit	Base unit

Table: Sizes and their base units

Size	Formula	Unit	Base unit
Force	F	Newton, N	1 N = 1 kg m/s ²

Table: Sizes and their base units

Force is mass times acceleration: $F = m \cdot a$

Size	Formula	Unit	Base unit
Force	F	Newton, N	$1 \text{ N} = 1 \text{ kg m/s}^2$
Energy	E	Joule, J	$1 \text{ J} = 1 \text{ Ws} = 1 \text{ Nm} = 1 \text{ kg m}^2/\text{s}^2$

Table: Sizes and their base units

Often comparable to work, important for energy conservation, unit also watt-second

Size	Formula	Unit	Base unit
Force	F	Newton, N	$1 \text{ N} = 1 \text{ kg m/s}^2$
Energy	E	Joule, J	$1 \text{ J} = 1 \text{ Ws} = 1 \text{ Nm} = 1 \text{ kg m}^2/\text{s}^2$
Power	P	Watt, W	$1 \text{ W} = 1 \text{ J/s} = 1 \text{ kg m}^2/\text{s}^3$

Table: Sizes and their base units

Power is work per time

Size	Formula	Unit	Base unit
Force	F	Newton, N	$1 \text{ N} = 1 \text{ kg m/s}^2$
Energy	E	Joule, J	$1 \text{ J} = 1 \text{ Ws} = 1 \text{ Nm} = 1 \text{ kg m}^2/\text{s}^2$
Power	P	Watt, W	$1 \text{ W} = 1 \text{ J/s} = 1 \text{ kg m}^2/\text{s}^3$
Voltage	U	Volt, V	$1 \text{ V} = 1 \text{ W/A} = 1 \text{ Nm/As} = 1 \text{ kg m}^2/\text{s}^3\text{A}$

Table: Sizes and their base units

“Strength” of a voltage source - analogous to pressure in a pipe

Size	Formula	Unit	Base unit
Force	F	Newton, N	$1 \text{ N} = 1 \text{ kg m/s}^2$
Energy	E	Joule, J	$1 \text{ J} = 1 \text{ Ws} = 1 \text{ Nm} = 1 \text{ kg m}^2/\text{s}^2$
Power	P	Watt, W	$1 \text{ W} = 1 \text{ J/s} = 1 \text{ kg m}^2/\text{s}^3$
Voltage	U	Volt, V	$1 \text{ V} = 1 \text{ W/A} = 1 \text{ Nm/As} = 1 \text{ kg m}^2/\text{s}^3\text{A}$
Load	Q	Coulomb, C	$1 \text{ C} = 1 \text{ As}$

Table: Sizes and their base units

Amount of free elementary charges

Size	Formula	Unit	Base unit
Force	F	Newton, N	$1 \text{ N} = 1 \text{ kg m/s}^2$
Energy	E	Joule, J	$1 \text{ J} = 1 \text{ Ws} = 1 \text{ Nm} = 1 \text{ kg m}^2/\text{s}^2$
Power	P	Watt, W	$1 \text{ W} = 1 \text{ J/s} = 1 \text{ kg m}^2/\text{s}^3$
Voltage	U	Volt, V	$1 \text{ V} = 1 \text{ W/A} = 1 \text{ Nm/As} = 1 \text{ kg m}^2/\text{s}^3\text{A}$
Load	Q	Coulomb, C	$1 \text{ C} = 1 \text{ As}$
Resistance	R	Ohm, Ω	$1 \Omega = 1 \text{ V/A} = 1 \text{ kg m}^2/\text{s}^3\text{A}^2$

Table: Sizes and their base units

“Friction” of electric current

Size	Formula	Unit	Base unit
Force	F	Newton, N	$1 \text{ N} = 1 \text{ kg m/s}^2$
Energy	E	Joule, J	$1 \text{ J} = 1 \text{ Ws} = 1 \text{ Nm} = 1 \text{ kg m}^2/\text{s}^2$
Power	P	Watt, W	$1 \text{ W} = 1 \text{ J/s} = 1 \text{ kg m}^2/\text{s}^3$
Voltage	U	Volt, V	$1 \text{ V} = 1 \text{ W/A} = 1 \text{ Nm/As} = 1 \text{ kg m}^2/\text{s}^3\text{A}$
Load	Q	Coulomb, C	$1 \text{ C} = 1 \text{ As}$
Resistance	R	Ohm, Ω	$1 \Omega = 1 \text{ V/A} = 1 \text{ kg m}^2/\text{s}^3\text{A}^2$
Capacity	C	Farad, F	$1 \text{ F} = 1 \text{ As/V} = 1 \text{ s}^4\text{A}^2/\text{kg m}^2$

Table: Sizes and their base units

Ability to store charges

Size	Formula	Unit	Base unit
Force	F	Newton, N	$1 \text{ N} = 1 \text{ kg m/s}^2$
Energy	E	Joule, J	$1 \text{ J} = 1 \text{ Ws} = 1 \text{ Nm} = 1 \text{ kg m}^2/\text{s}^2$
Power	P	Watt, W	$1 \text{ W} = 1 \text{ J/s} = 1 \text{ kg m}^2/\text{s}^3$
Voltage	U	Volt, V	$1 \text{ V} = 1 \text{ W/A} = 1 \text{ Nm/As} = 1 \text{ kg m}^2/\text{s}^3\text{A}$
Load	Q	Coulomb, C	$1 \text{ C} = 1 \text{ As}$
Resistance	R	Ohm, Ω	$1 \Omega = 1 \text{ V/A} = 1 \text{ kg m}^2/\text{s}^3\text{A}^2$
Capacity	C	Farad, F	$1 \text{ F} = 1 \text{ As/V} = 1 \text{ s}^4\text{A}^2/\text{kg m}^2$
Inductance	L	Henry, H	$1 \text{ H} = 1 \text{ Vs/A} = 1 \text{ kg m}^2/\text{s}^2\text{A}^2$

Table: Sizes and their base units

Key characteristic of an electric coil

Size	Formula	Unit	Base unit
Force	F	Newton, N	$1 \text{ N} = 1 \text{ kg m/s}^2$
Energy	E	Joule, J	$1 \text{ J} = 1 \text{ Ws} = 1 \text{ Nm} = 1 \text{ kg m}^2/\text{s}^2$
Power	P	Watt, W	$1 \text{ W} = 1 \text{ J/s} = 1 \text{ kg m}^2/\text{s}^3$
Voltage	U	Volt, V	$1 \text{ V} = 1 \text{ W/A} = 1 \text{ Nm/As} = 1 \text{ kg m}^2/\text{s}^3\text{A}$
Load	Q	Coulomb, C	$1 \text{ C} = 1 \text{ As}$
Resistance	R	Ohm, Ω	$1 \Omega = 1 \text{ V/A} = 1 \text{ kg m}^2/\text{s}^3\text{A}^2$
Capacity	C	Farad, F	$1 \text{ F} = 1 \text{ As/V} = 1 \text{ s}^4\text{A}^2/\text{kg m}^2$
Inductance	L	Henry, H	$1 \text{ H} = 1 \text{ Vs/A} = 1 \text{ kg m}^2/\text{s}^2\text{A}^2$
Magn. Flux	Φ	Weber, Wb	$1 \text{ Wb} = 1 \text{ Vs} = 1 \text{ kg m}^2/\text{s}^2\text{A}$

Table: Sizes and their base units

Analogous property of a magnetic field to electric current

Size	Formula	Unit	Base unit
Force	F	Newton, N	$1 \text{ N} = 1 \text{ kg m/s}^2$
Energy	E	Joule, J	$1 \text{ J} = 1 \text{ Ws} = 1 \text{ Nm} = 1 \text{ kg m}^2/\text{s}^2$
Power	P	Watt, W	$1 \text{ W} = 1 \text{ J/s} = 1 \text{ kg m}^2/\text{s}^3$
Voltage	U	Volt, V	$1 \text{ V} = 1 \text{ W/A} = 1 \text{ Nm/As} = 1 \text{ kg m}^2/\text{s}^3\text{A}$
Load	Q	Coulomb, C	$1 \text{ C} = 1 \text{ As}$
Resistance	R	Ohm, Ω	$1 \Omega = 1 \text{ V/A} = 1 \text{ kg m}^2/\text{s}^3\text{A}^2$
Capacity	C	Farad, F	$1 \text{ F} = 1 \text{ As/V} = 1 \text{ s}^4\text{A}^2/\text{kg m}^2$
Inductance	L	Henry, H	$1 \text{ H} = 1 \text{ Vs/A} = 1 \text{ kg m}^2/\text{s}^2\text{A}^2$
Magn. Flux	Φ	Weber, Wb	$1 \text{ Wb} = 1 \text{ Vs} = 1 \text{ kg m}^2/\text{s}^2\text{A}$
Flux density	B	Tesla, T	$1 \text{ T} = 1 \text{ Vs/m}^2 = 1 \text{ kg/s}^2\text{A}$

Table: Sizes and their base units

Indicates the strength of the magnetic flux, e.g., in MRI

Designation	Potency	Potency	Designation

Designation	Potency	Potency	Designation
Dezi, d	10^{-1}	10^1	Deka, da

Designation	Potency	Potency	Designation
Dezi, d	10^{-1}	10^1	Deka, da
Zenti, c	10^{-2}	10^2	Hekto, h

Designation	Potency	Potency	Designation
Dezi, d	10^{-1}	10^1	Deka, da
Zenti, c	10^{-2}	10^2	Hekto, h
Milli, m	10^{-3}	10^3	Kilo, k

Designation	Potency	Potency	Designation
Dezi, d	10^{-1}	10^1	Deka, da
Zenti, c	10^{-2}	10^2	Hekto, h
Milli, m	10^{-3}	10^3	Kilo, k
Mikro, μ	10^{-6}	10^6	Mega, M

Designation	Potency	Potency	Designation
Dezi, d	10^{-1}	10^1	Deka, da
Zenti, c	10^{-2}	10^2	Hekto, h
Milli, m	10^{-3}	10^3	Kilo, k
Mikro, μ	10^{-6}	10^6	Mega, M
Nano, n	10^{-9}	10^9	Giga, G

Designation	Potency	Potency	Designation
Dezi, d	10^{-1}	10^1	Deka, da
Zenti, c	10^{-2}	10^2	Hekto, h
Milli, m	10^{-3}	10^3	Kilo, k
Mikro, μ	10^{-6}	10^6	Mega, M
Nano, n	10^{-9}	10^9	Giga, G
Piko, p	10^{-12}	10^{12}	Tera, T

Designation	Potency	Potency	Designation
Dezi, d	10^{-1}	10^1	Deka, da
Zenti, c	10^{-2}	10^2	Hekto, h
Milli, m	10^{-3}	10^3	Kilo, k
Mikro, μ	10^{-6}	10^6	Mega, M
Nano, n	10^{-9}	10^9	Giga, G
Piko, p	10^{-12}	10^{12}	Tera, T
Femto, f	10^{-15}	10^{15}	Peta, P

Designation	Potency	Potency	Designation
Dezi, d	10^{-1}	10^1	Deka, da
Zenti, c	10^{-2}	10^2	Hekto, h
Milli, m	10^{-3}	10^3	Kilo, k
Mikro, μ	10^{-6}	10^6	Mega, M
Nano, n	10^{-9}	10^9	Giga, G
Piko, p	10^{-12}	10^{12}	Tera, T
Femto, f	10^{-15}	10^{15}	Peta, P
Atto, a	10^{-18}	10^{18}	Exa, E

Designation	Potency	Potency	Designation
Dezi, d	10^{-1}	10^1	Deka, da
Zenti, c	10^{-2}	10^2	Hekto, h
Milli, m	10^{-3}	10^3	Kilo, k
Mikro, μ	10^{-6}	10^6	Mega, M
Nano, n	10^{-9}	10^9	Giga, G
Piko, p	10^{-12}	10^{12}	Tera, T
Femto, f	10^{-15}	10^{15}	Peta, P
Atto, a	10^{-18}	10^{18}	Exa, E

$$1 \text{ m} = 100 \cdot 10^{-2} \text{ m} = 100 \text{ cm}$$

Designation	Potency	Potency	Designation
Dezi, d	10^{-1}	10^1	Deka, da
Zenti, c	10^{-2}	10^2	Hekto, h
Milli, m	10^{-3}	10^3	Kilo, k
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Piko, p	10^{-12}	10^{12}	Tera, T
Femto, f	10^{-15}	10^{15}	Peta, P
Atto, a	10^{-18}	10^{18}	Exa, E

$$1 \text{ m} = 100 \cdot 10^{-2} \text{ m} = 100 \text{ cm}$$

$$1 \text{ m}^2 = 100^2 \cdot (10^{-2} \text{ m})^2 = 10.000 \text{ cm}^2$$

Designation	Potency	Potency	Designation
Dezi, d	10^{-1}	10^1	Deka, da
Zenti, c	10^{-2}	10^2	Hekto, h
Milli, m	10^{-3}	10^3	Kilo, k
Mikro, μ	10^{-6}	10^6	Mega, M
Nano, n	10^{-9}	10^9	Giga, G
Piko, p	10^{-12}	10^{12}	Tera, T
Femto, f	10^{-15}	10^{15}	Peta, P
Atto, a	10^{-18}	10^{18}	Exa, E

$$1 \text{ m} = 100 \cdot 10^{-2} \text{ m} = 100 \text{ cm}$$

$$1 \text{ m}^2 = 100^2 \cdot (10^{-2} \text{ m})^2 = 10.000 \text{ cm}^2$$

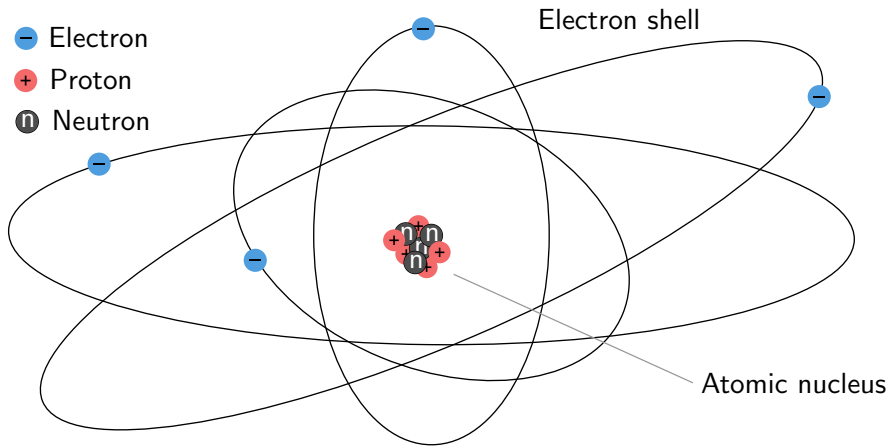
$$1 \text{ m}^3 = 100^3 \cdot (10^{-2} \text{ m})^3 = 1.000.000 \text{ cm}^3$$

Learning objectives: The electrical charge

The students can

- ▶ describe the properties of electric charges and related physical phenomena
- ▶ describe electric fields and calculate them for simple charge configurations
- ▶ Calculate forces on charges using Coulomb's law

Atomic structure - Bohr's atomic model



The smallest unit of charge is the elementary charge:

$$e = 1,6 \cdot 10^{-19} \text{ C (Natural constant!)}$$

Key point:

$$Q = \pm n \cdot e$$

with $n = 1, 2, 3, \dots$

$$[Q] = 1 \text{ Coulomb} = 1 \text{ C} = 1 \text{ As}$$

Electrons and ions

The smallest unit of charge is the elementary charge:

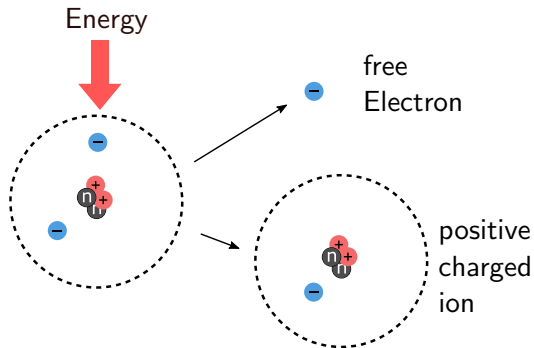
$$e = 1,6 \cdot 10^{-19} \text{ C (Natural constant!)}$$

Key point:

$$Q = \pm n \cdot e$$

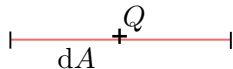
with $n = 1, 2, 3, \dots$

$$[Q] = 1 \text{ Coulomb} = 1 \text{ C} = 1 \text{ As}$$



Free electrons are significant as the smallest, lightest, and most mobile charge carriers!

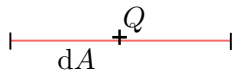
line load



$$\lambda = \lim_{l \rightarrow 0} \frac{\Delta Q}{\Delta l} = \frac{dQ}{dl}$$

$$Q = \int_l \lambda dl$$

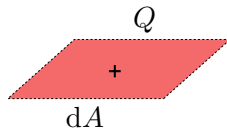
line load



$$\lambda = \lim_{l \rightarrow 0} \frac{\Delta Q}{\Delta l} = \frac{dQ}{dl}$$

$$Q = \int_l \lambda dl$$

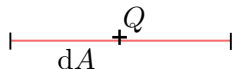
Surface charge



$$\sigma = \lim_{A \rightarrow 0} \frac{\Delta Q}{\Delta A} = \frac{dQ}{dA}$$

$$Q = \iint_A \sigma dA$$

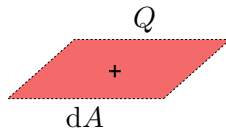
line load



$$\lambda = \lim_{l \rightarrow 0} \frac{\Delta Q}{\Delta l} = \frac{dQ}{dl}$$

$$Q = \int_l \lambda dl$$

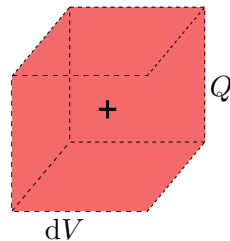
Surface charge



$$\sigma = \lim_{A \rightarrow 0} \frac{\Delta Q}{\Delta A} = \frac{dQ}{dA}$$

$$Q = \iint_A \sigma dA$$

Space charge



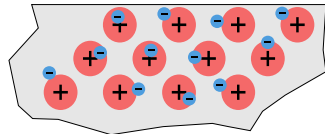
$$\rho = \lim_{V \rightarrow 0} \frac{\Delta Q}{\Delta V} = \frac{dQ}{dV}$$

$$Q = \iiint_V \rho dV$$

Electrical conductors - metals

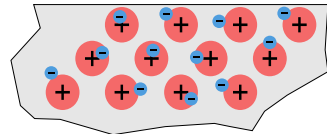
Atomic arrangement in **Lattice structure**,
outer electrons almost freely movable
→ gives electrically conductive properties

Total electrons $\hat{=}$ Total protons
→ electrically neutral



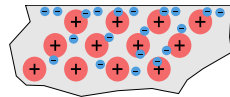
Electrical conductors - metals

Atomic arrangement in **Lattice structure**,
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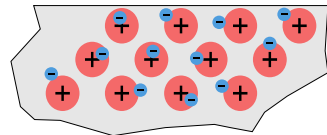
Total electrons $\hat{=}$ Total protons
→ electrically neutral

Adding electrons
→ negatively charged



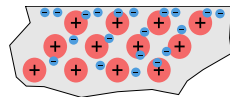
Electrical conductors - metals

Atomic arrangement in **Lattice structure**,
outer electrons almost freely movable
→ gives electrically conductive properties

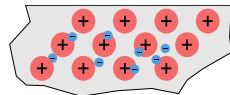


Total electrons $\hat{=}$ Total protons
→ electrically neutral

Adding electrons
→ negatively charged



Removal of electrons
→ positively charged

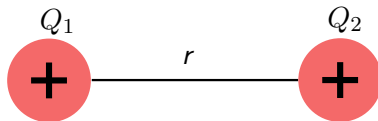


Coulomb's law

Charges exert forces on each other.

Like charge $\rightarrow$ repulsive

Unequal charge $\rightarrow$ attractive

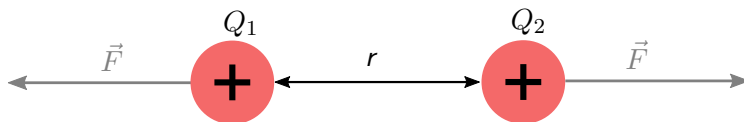


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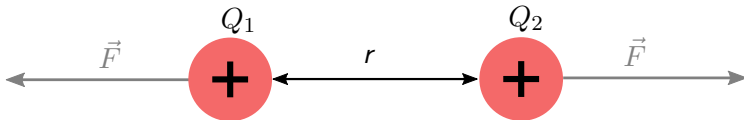


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Like charge $\rightarrow$ repulsive

Unequal charge $\rightarrow$ attractive



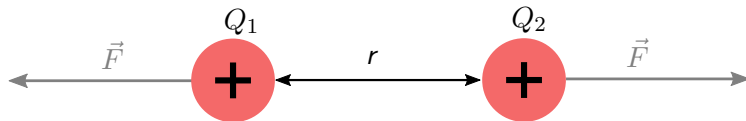
$$F \sim \frac{Q_1 Q_2}{r^2} \rightarrow F = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{r^2}$$

Coulomb's law

Charges exert forces on each other.

Like charge → repulsive

Unequal charge → attractive



$$F \sim \frac{Q_1 Q_2}{r^2} \rightarrow F = \frac{1}{4\pi\epsilon_0} \frac{Q_1 Q_2}{r^2}$$

ϵ_0 : **electric field constant (dielectric constant)**

$$\epsilon_0 = 8,854 \cdot 10^{-12} \text{As/Vm}$$

Learning objectives: The electric field

The students can

- ▶ develop a 'feel' for electric fields
- ▶ describe electric fields and calculate them for simple charge configurations
- ▶ characterise the behaviour of electric fields on conductors

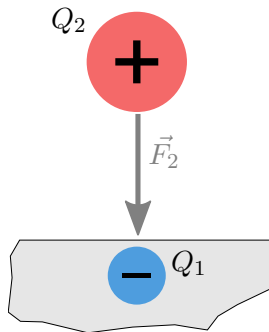
Characterisation of the electric field

Electric field $\vec{E}$ defined by:

Size (in Newton)

Direction

of a **Force** $\vec{F}_E$ on pos. trial load



Characterisation of the electric field

Electric field $\vec{E}$ defined by:

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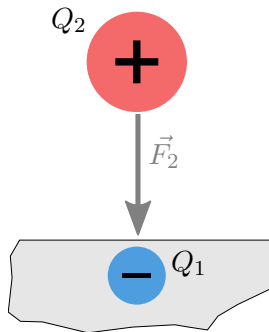
of a **Force** $\vec{F}_E$ on pos. trial load

Characterized by **field lines**

→ Vector field

Disadvantage:

Field strength **independent** of sample charge



Characterisation of the electric field

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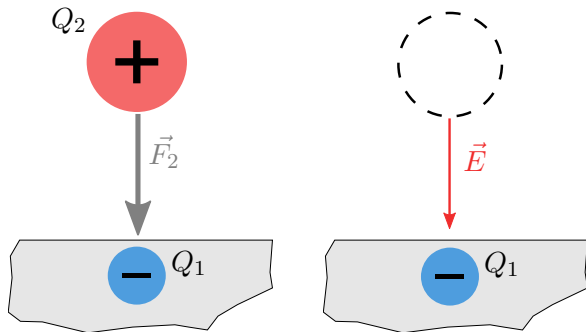
→ Vector field

Disadvantage:

Field strength **independent** of sample charge

Key point:

$$\vec{E}_1 = \frac{\vec{F}_2}{Q_2}$$



Characterisation of the electric field

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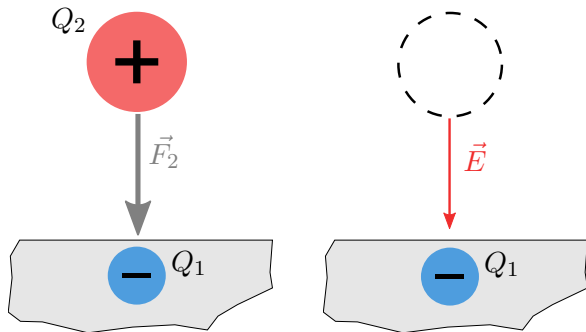
→ Vector field

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Field strength **independent** of sample charge

Key point:

$$\vec{E}_1 = \frac{\vec{F}_2}{Q_2}$$



$$\vec{E}_1 = \frac{\vec{F}_2}{Q_2} = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q_1 \cdot Q_2}{r^2} \cdot \frac{1}{Q_2} = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q_1}{r^2}$$

Electrical conductors and electrostatic fields I

Field lines always begin and end at charges (often conductor surfaces)..

Electron surplus:

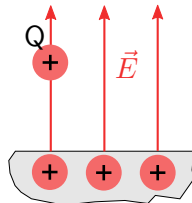
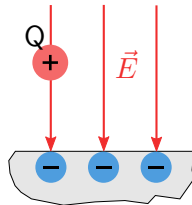
- Pos. Test charge is attracted
- Field lines point toward the conductor

Electron deficiency:

- Pos. Test charge is repelled
- Field lines point away from the conductor

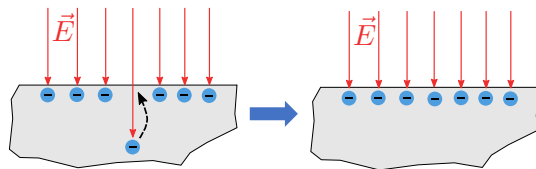
Key point:

In conductors, the electrostatic field disappears.



Conductors and electrostatic fields II

Movable charge carriers: $\rightarrow$ conductor surface $\rightarrow$ electrostatic field is compensated



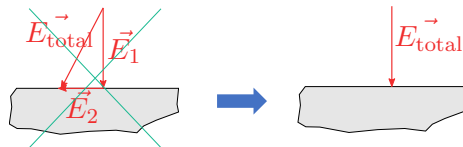
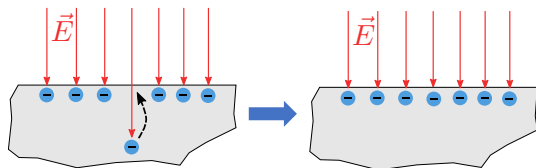
Conductors and electrostatic fields II

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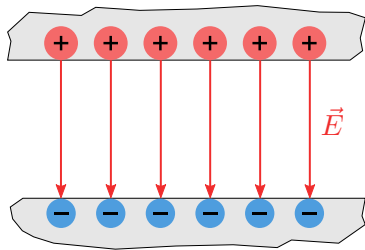
Vector decomposition of $\vec{E}$
 $\rightarrow$ Horizontal component eliminated

Key point:

Electric field lines are perpendicular to conductors.

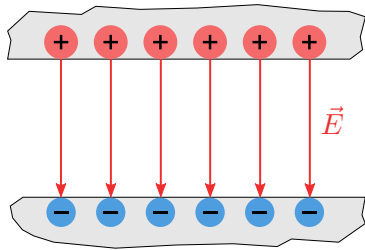


Examples of electric fields I

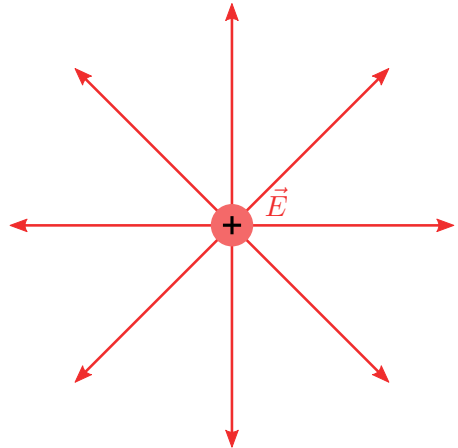


Electric homogeneous field between two
printed circuit boards

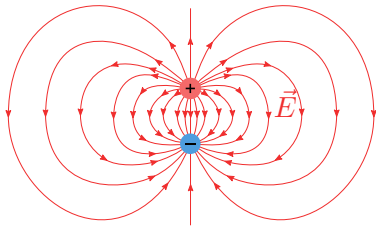
Examples of electric fields I



Electric homogeneous field between two printed circuit boards

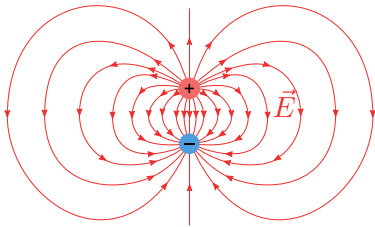


Radial field of a positive point charge

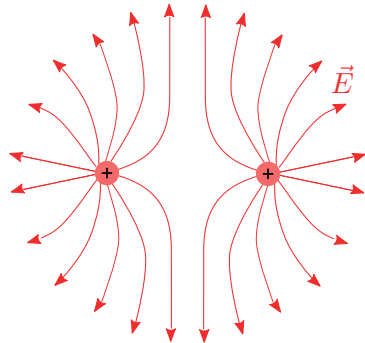


Electric field of attracting point charges

Examples of electric fields II



Electric field of attracting point charges



Electric field of repelling point charges

Learning objectives: The electrical potential

The Students can

- ▶ Calculate displacement work of charges in an electric field
- ▶ Determine and draw equipotential surfaces
- ▶ determine electrical voltages from given field sizes

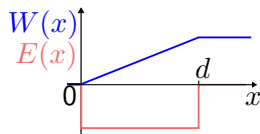
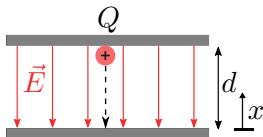
Working in an electric field

Work required to move charges in the electric field

Work $W \triangleq \text{Force} \cdot \text{Distance}$
 $[W] = \text{J (Joule)}$

Sign depends on the observer; here from the producer's perspective:

Negative work $\rightarrow$ Increase in potential energy in the system



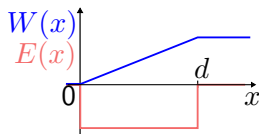
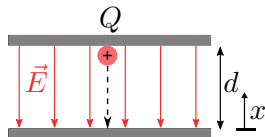
Working in an electric field

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Work $W \hat{=}$ Force $\cdot$ Distance
 $[W] = \text{J (Joule)}$

Sign depends on the observer; here from the producer's perspective:

Negative work $\rightarrow$ Increase in potential energy in the system



$$W = \int_d^0 \vec{F} d\vec{x} = Q \int_d^0 \vec{E} \cdot d\vec{x}$$

$$\text{with } \vec{E} = \frac{\vec{F}}{Q}$$

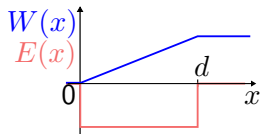
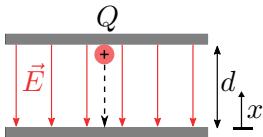
$$W = -Q \cdot E \int_d^0 dx = -Q \cdot E [x]_d^0$$

$$W = -Q \cdot E(0 - d) = Q \cdot E \cdot d$$

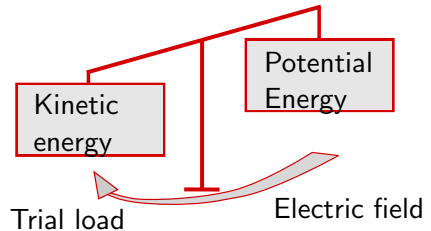
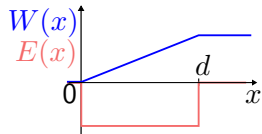
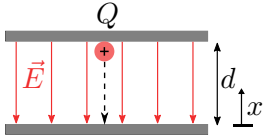
Key point:

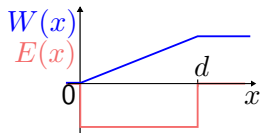
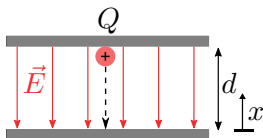
$$\Delta W = Q \cdot E \cdot \Delta x$$

Working in the electrical field II

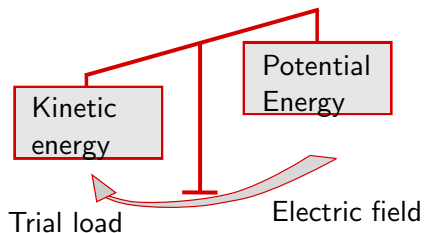


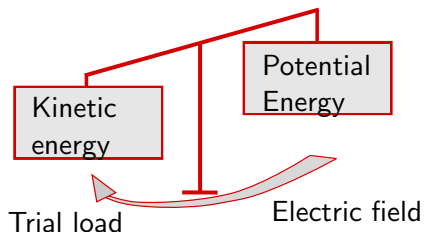
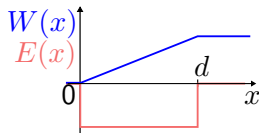
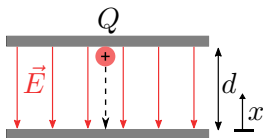
Working in the electrical field II





$$W(x) = Q \cdot E \cdot x + W_0$$





$$W(x) = Q \cdot E \cdot x + W_0$$

Problem: Dependence on trial loading Q

→ **Normierung** from $W(x)$ to Q as work potential

$$\frac{W(x)}{Q} = E \cdot x + \frac{W_0}{Q}$$

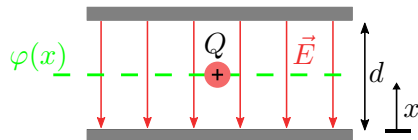
Definition des elektrischen Potentials

Standardization of potential energy:

$$\text{Potential } \varphi = \frac{W}{Q}$$

$$[\varphi] = \text{Volt} = \text{V}$$

In an electric homogeneous field:



$$\varphi(x) = E \cdot x + \varphi_0$$

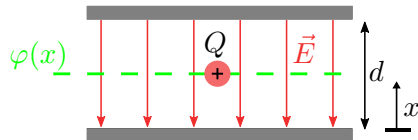
Definition des elektrischen Potentials

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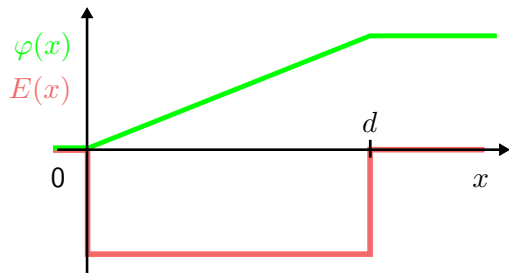
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In an electric homogeneous field:



$$\varphi(x) = E \cdot x + \varphi_0$$



$$E = -\frac{\Delta\varphi}{\Delta x}$$

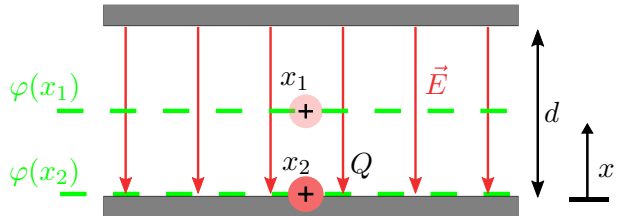
Reference point required:

$$\varphi_0 = \varphi(x = 0) = 0$$

Connection between work and potential

Potential only dependent on local coordinate x in field direction.

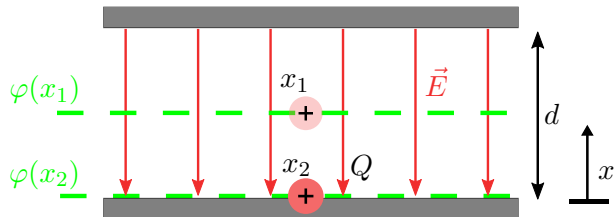
Surfaces of equal potential: →
Equipotential surfaces



Connection between work and potential

Potential only dependent on local coordinate x in field direction.

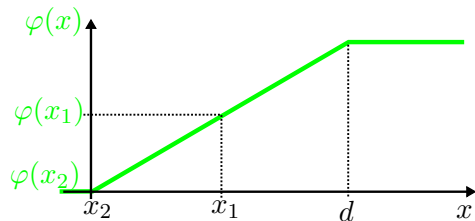
Surfaces of equal potential: →
Equipotential surfaces



$$W_{12} = Q \cdot (\varphi_1 - \varphi_2)$$

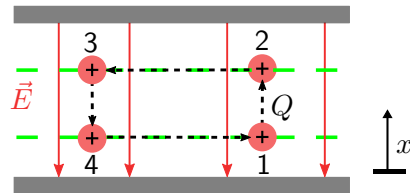
With reference potential $\varphi_2 = 0$:

$$W_{12} = Q \cdot (\varphi_1 - 0) = Q \cdot \varphi_1$$



Vortice-free field

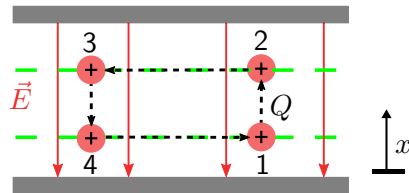
First, consideration of the partial energies in a closed cycle in a homogeneous field.



Vortice-free field

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The total energy is calculated by adding up the partial energies:



Vortice-free field

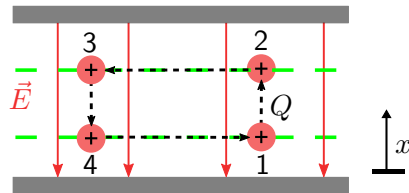
First, consideration of the partial energies in a closed cycle in a homogeneous field.

The total energy is calculated by adding up the partial energies:

$$W_{\text{ges}} = Q \cdot (\varphi_1 - \varphi_2) + Q \cdot (\varphi_2 - \varphi_3) \\ + Q \cdot (\varphi_3 - \varphi_4) + Q \cdot (\varphi_4 - \varphi_1)$$

$$W_{\text{ges}} = Q \cdot (\varphi_1 - \varphi_2 + \varphi_2 - \varphi_3 + \varphi_3 - \varphi_4 + \varphi_4 - \varphi_1)$$

$$W_{\text{ges}} = Q \cdot 0 = 0$$



Vortice-free field

First, consideration of the partial energies in a closed cycle in a homogeneous field.

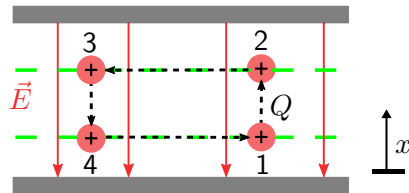
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$$W_{\text{ges}} = Q \cdot 0 = 0$$

Energy in the electric field depends only on the starting and ending points, not on the path. It is a “vortex-free” field.



Marking:

- ▶ no closed field lines
- ▶ Field lines begin and end at charges

Definition of electrical voltage

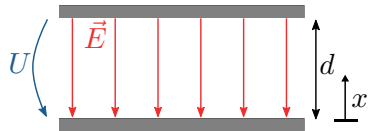
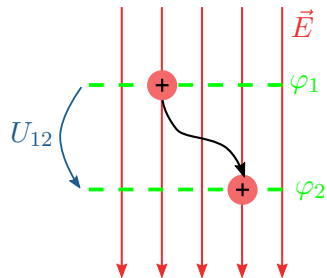
Potential difference determines usable energy

→ **Voltage** U :

$$U_{12} = (\varphi_1 - \varphi_2)$$

$$[U] = \text{Volt} = \text{V}$$

Index order: Start and end point



Relationship between voltage and field strength I

In **Homogeneous field**:

$$\varphi(x) = E \cdot x$$

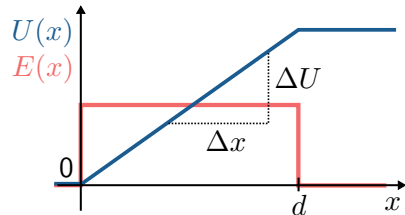
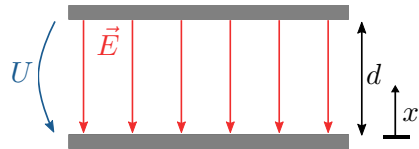
$$U = \varphi(x = d) - \varphi(x = 0)$$

$$U = E \cdot d - E \cdot 0$$

Key point:

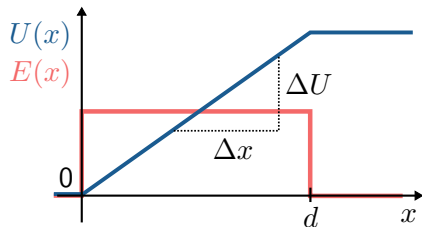
$$U = E \cdot d$$

$$E = \frac{U}{d}$$



Relationship between voltage and field strength II

Voltage $\hat{=}$ integrated field strength:

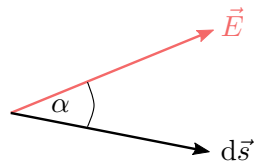


General field:

Key point:

$$U_{12} = \int_1^2 \vec{E} \, d\vec{s}$$

Repetition of scalar product:



$$\vec{E} \cdot d\vec{s} = E \, ds \cdot \cos(\alpha)$$

Learning objectives: The electrical current

The Students can

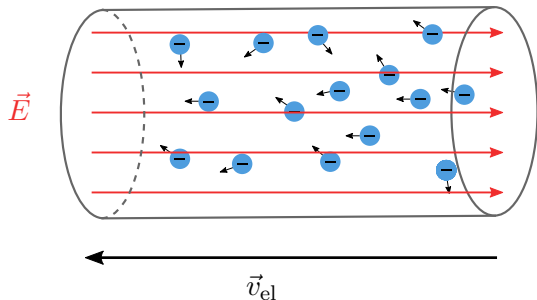
- ▶ Explain and apply the concepts of electric current and electric current density. Apply
- ▶ calculate the electric current and electric current density in simple arrangements
- ▶ determine the drift velocity of electrons in simple arrangements

The electric flow field

Flow field: **directed** motion of particles

Stationary flow field: **time-constant**
particle flow

Stationary electric flow field: flowing
particles **electric charge carriers**



Electric current I

Electric current: directed drift motion of charge carriers

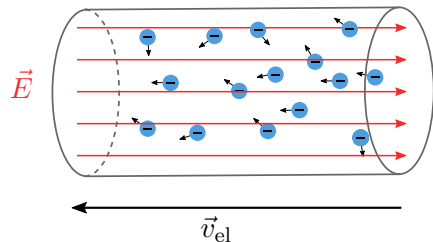
Field strength-dependent drift velocity $\vec{v}_{\text{el}}$:

$$\vec{v}_{\text{el}} = -b_{\text{el}} \cdot \vec{E}$$

with:

$\vec{v}_{\text{el}}$ = Drift velocity of electrons in m/s

b_{el} = Electron mobility in cm^2/Vs



Charge carrier transport within a conductor:

$$\Delta Q = e \cdot N_{\text{el}}$$

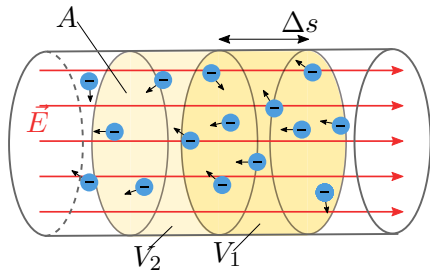
$$\Delta Q = e \cdot n_{\text{el}} \cdot V_1$$

$$\Delta Q = e \cdot n_{\text{el}} \cdot b_{\text{el}} \cdot E \cdot \Delta t \cdot A$$

with:

N_{el} = Number of load carriers in the volume under consideration

n_{el} = Charge carrier density in $1/\text{m}^3$



Definition of electric current

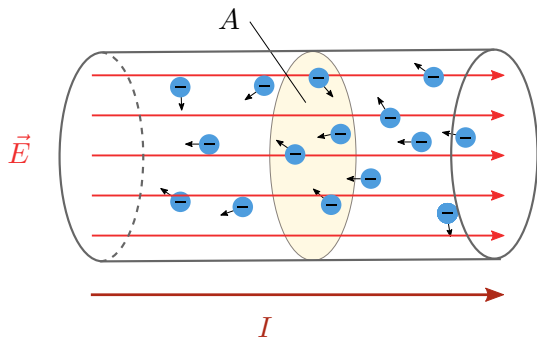
Current I : Amount of charge ΔQ in Δt through a conductor

Key point:

$$I = \frac{\Delta Q}{\Delta t}$$

$$[I] = \text{Ampere} = \text{A}$$

$$I = e \cdot n_{\text{el}} \cdot b_{\text{el}} \cdot E \cdot A$$



Definition of electric current density

Current J : Current ΔI per conductor cross-section ΔA

Key point:

$$J = \frac{\Delta I}{\Delta A}$$

$$[J] = \frac{\text{A}}{\text{m}^2}$$

$$J_1 \cdot A_1 = J_2 \cdot A_2 = I$$

